INSTITUT NATIONAL
DE L'INFORMATION
GÉOGRAPHIQUE
ET FORESTIÈRE

Metrology with MicMac v2

JR IGN - 2025

JM Muller

Coded targets

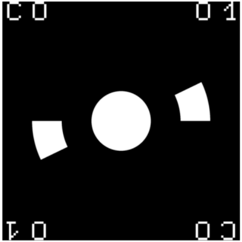
Topometry

Simulation

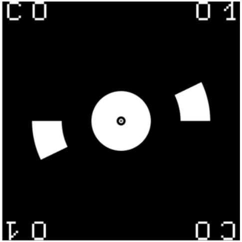
Conclusion

Coded targets

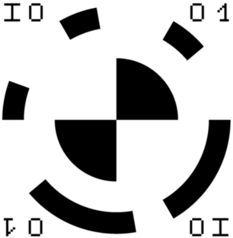
Targets types



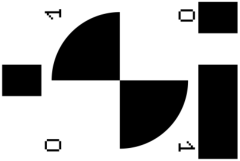
"Circular"



"Circular Topo"

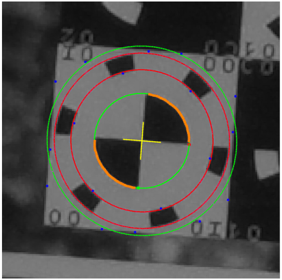


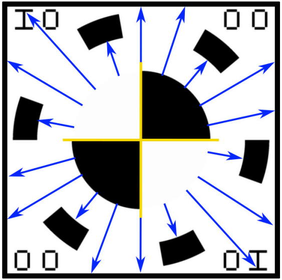
"IGN Indoor"



"IGN Aerial"

Detection / decoding

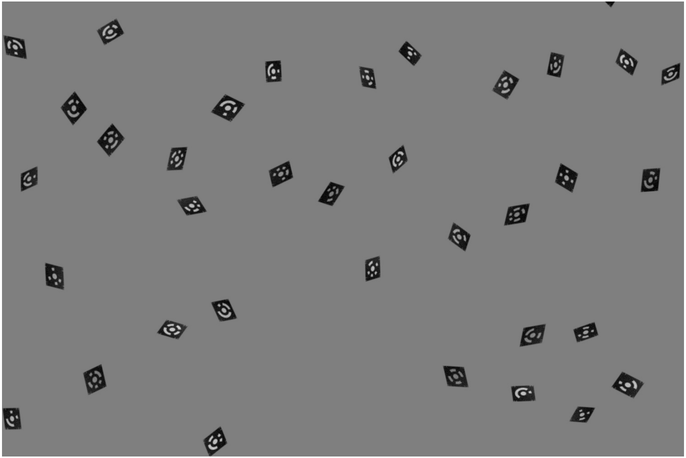




Paper

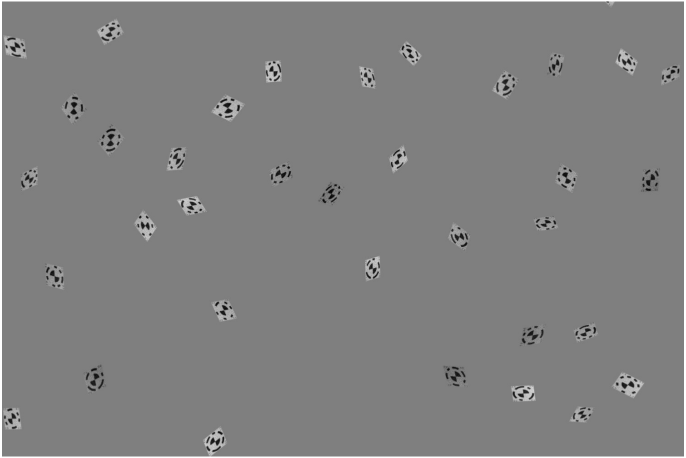
Muller, J.-M., Daakir, M., Pierrot Deseilligny, M., and Barcet, F.: **Implementation of coded targets for metrology applications in MicMac, a free open-source photogrammetric software**, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLVIII-2/W7-2024, 89–96, doi.org/10.5194/isprs-archives-XLVIII-2-W7-2024-89-2024, 2024.

2D simulations

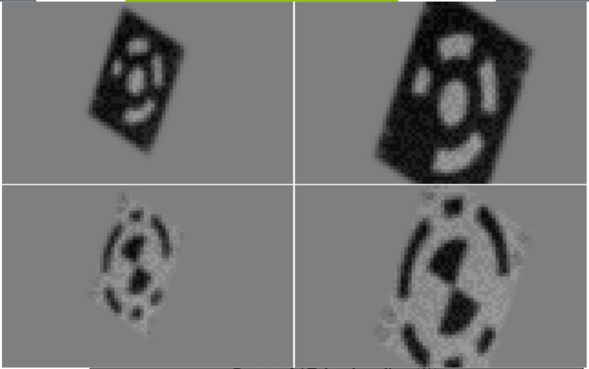


Tested softwares: DPA Pro, CERN, MMVII

2D simulations



Tested software: MMVII

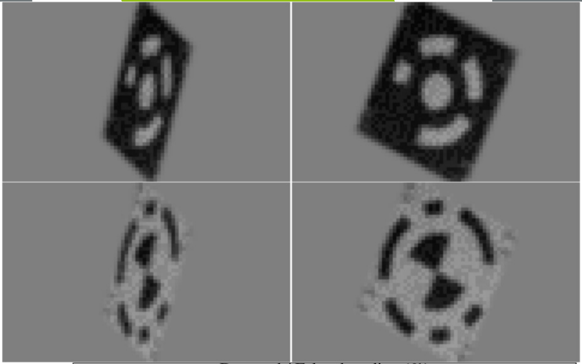


Target Size (px)	Detected / False decoding (%) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼30	34.4 / 0.0	1.5 / 0.3	33.0 / 4.8	95.0 / 0.1
∼35	52.2 / 0.0	8.5 / 0.6	64.5 / 3.5	97.8 / 0.1
∼40	52.4 / 0.0	31.5 / 1.1	86.5 / 0.7	99.2 / 0.0
∼45	51.4 / 0.0	58.1 / 2.6	95.5 / 0.1	99.8 / 0.0
∼50	43.8 / 0.1	79.9 / 2.8	97.7 / 0.0	100.0 / 0.0

(a) 2D simulation detection results for target size variation

Target Size (px)	Mean error ± sigma (px) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼30	0.06 ± 0.04	0.06 ± 0.04	0.05 ± 0.03	0.06 ± 0.03
∼35	0.07 ± 0.05	0.06 ± 0.03	0.05 ± 0.03	0.06 ± 0.03
∼40	0.06 ± 0.05	0.06 ± 0.04	0.05 ± 0.03	0.05 ± 0.03
∼45	0.05 ± 0.04	0.06 ± 0.04	0.05 ± 0.02	0.05 ± 0.03
∼50	0.05 ± 0.04	0.07 ± 0.04	0.05 ± 0.02	0.05 ± 0.02

(b) 2D simulation center detection results for target size variation

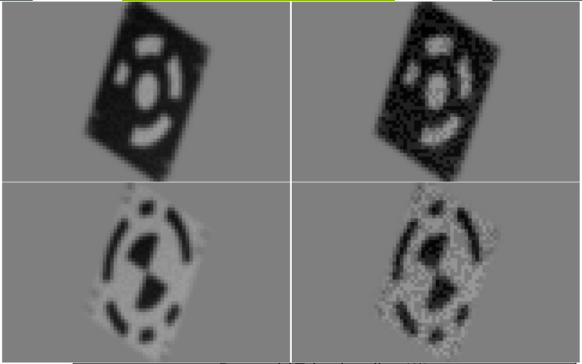


Target Ratio	Detected / False decoding (%) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼0.5	33.0 / 0.0	1.9 / 0.1	10.0 / 0.6	88.0 / 0.0
∼0.6	48.4 / 0.0	12.5 / 0.3	47.9 / 1.1	96.8 / 0.0
∼0.7	52.4 / 0.0	31.5 / 1.1	86.5 / 0.7	99.2 / 0.0
∼0.8	52.8 / 0.0	50.9 / 3.5	98.0 / 0.5	99.8 / 0.0
∼0.9	51.7 / 0.0	61.8 / 4.2	99.2 / 0.1	100.0 / 0.0

(a) 2D simulation detection results for target ratio variation

Target Ratio	Mean error ± sigma (px) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼0.5	0.08 ± 0.05	0.08 ± 0.04	0.05 ± 0.03	0.06 ± 0.03
∼0.6	0.07 ± 0.05	0.07 ± 0.04	0.05 ± 0.03	0.06 ± 0.03
∼0.7	0.06 ± 0.05	0.06 ± 0.04	0.05 ± 0.03	0.05 ± 0.03
∼0.8	0.06 ± 0.05	0.06 ± 0.04	0.05 ± 0.03	0.05 ± 0.02
∼0.9	0.05 ± 0.05	0.07 ± 0.04	0.05 ± 0.02	0.05 ± 0.02

(b) 2D simulation center detection results for target ratio variation

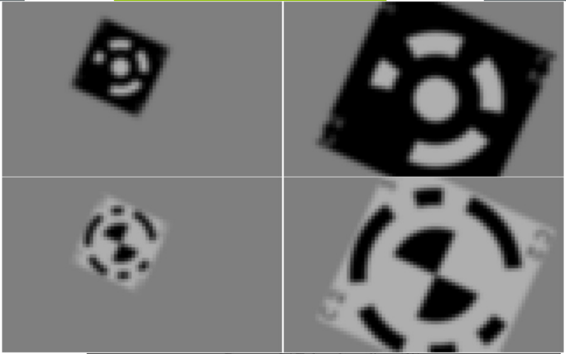


Target Noise	Detected / False decoding (%) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼0.02	96.5 / 0.0	63.5 / 1.4	97.9 / 0.3	100.0 / 0.0
∼0.04	93.2 / 0.0	49.6 / 1.5	95.5 / 0.7	99.8 / 0.0
∼0.06	52.4 / 0.0	31.5 / 1.1	86.5 / 0.7	99.2 / 0.0
∼0.08	16.2 / 0.0	13.0 / 0.8	57.3 / 1.1	97.7 / 0.0
∼0.10	5.5 / 0.0	3.7 / 0.2	23.3 / 1.4	94.4 / 0.0

(a) 2D simulation detection results for target noise variation

Target Noise	Mean error ± sigma (px) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼0.02	0.02 ± 0.02	0.05 ± 0.03	0.04 ± 0.01	0.05 ± 0.01
∼0.04	0.04 ± 0.03	0.05 ± 0.03	0.04 ± 0.02	0.05 ± 0.02
∼0.06	0.06 ± 0.05	0.06 ± 0.04	0.05 ± 0.03	0.05 ± 0.03
∼0.08	0.07 ± 0.05	0.08 ± 0.05	0.06 ± 0.03	0.06 ± 0.03
∼0.10	0.09 ± 0.06	0.09 ± 0.06	0.07 ± 0.04	0.07 ± 0.04

(b) 2D simulation center detection results for target noise variation



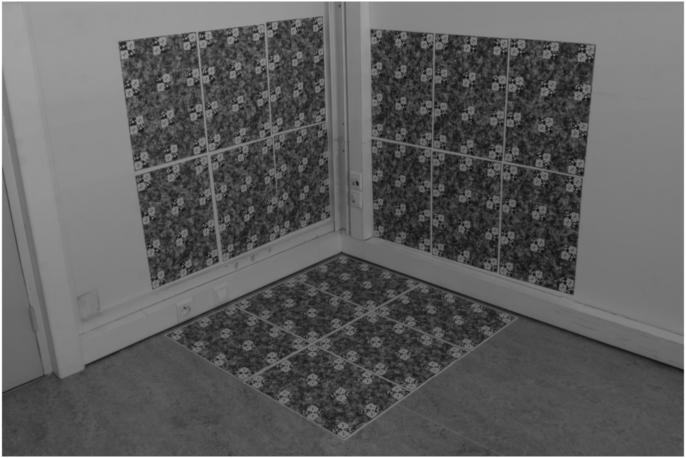
Target Size (px)	Detected / False decoding (%) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼20	7.7 / 0.0	2.0 / 0.2	29.3 / 9.1	98.3 / 0.0
∼30	89.2 / 0.0	54.7 / 0.3	96.0 / 9.7	100.0 / 0.0
∼35	94.5 / 0.0	88.9 / 0.8	99.8 / 0.5	100.0 / 0.0
∼40	99.3 / 0.0	97.9 / 1.1	100.0 / 0.0	100.0 / 0.0
∼45	99.5 / 0.0	98.0 / 1.4	100.0 / 0.0	100.0 / 0.0
∼50	99.7 / 0.0	99.8 / 0.6	100.0 / 0.0	100.0 / 0.0

(a) 2D simulation detection results for "ideal" target

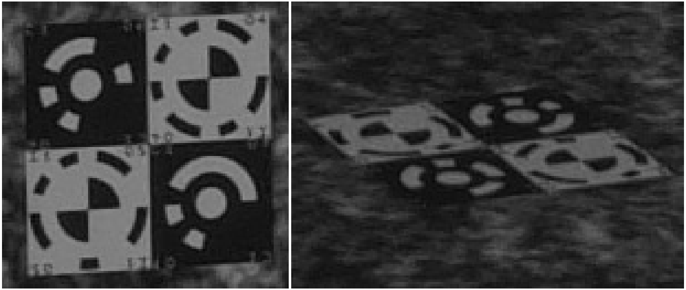
Target Size (px)	Mean error ± sigma (px) DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
∼20	0.033 ± 0.017	0.035 ± 0.025	0.0014 ± 0.0008	0.0014 ± 0.0008
∼30	0.014 ± 0.009	0.038 ± 0.022	0.0010 ± 0.0006	0.0015 ± 0.0006
∼35	0.010 ± 0.005	0.042 ± 0.024	0.0009 ± 0.0005	0.0016 ± 0.0006
∼40	0.010 ± 0.012	0.043 ± 0.024	0.0009 ± 0.0005	0.0018 ± 0.0006
∼45	0.010 ± 0.011	0.043 ± 0.025	0.0008 ± 0.0005	0.0021 ± 0.0006
∼50	0.010 ± 0.005	0.044 ± 0.025	0.0008 ± 0.0004	0.0023 ± 0.0005

(b) 2D simulation center detection results for "ideal" target

Real images

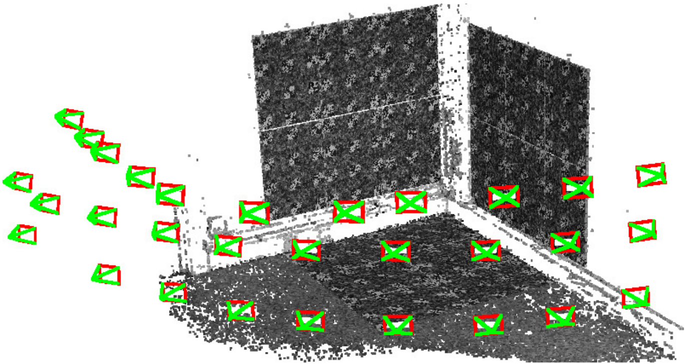


Real images



55x55 to 50x20 pixels targets

Real images



Real images

Detected / False decoding (%)			
DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
67.9 / 0.01	79.0 / 0.02	74.0 / 0.0	98.4 / 0.0

(a) Real-case detection results

Mean error ± sigma (px)			
DPA Pro Circular Target	CERN Circular Target	MMVII Circular Target	MMVII "IGN" Target
0.050 ± 0.033	0.072 ± 0.055	0.036 ± 0.024	0.043 ± 0.027

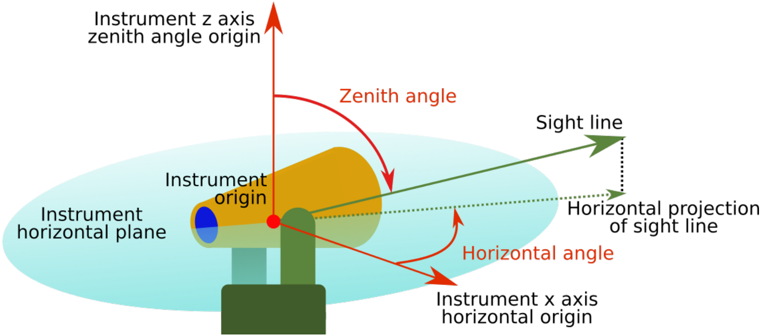
(b) Real-case center detection results

Topometry

Measurements

Station-based survey measurements are made from an instrument that is verticalized/plumb or not.

All the measurements are expressed in the station frame.



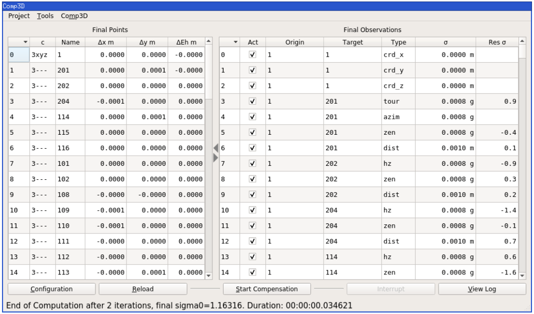
Measurements

The following measurements types are currently supported:

- Distances
- Horizontal angles
- Zenithal angles
- Direct Euclidian vectors
- Height differences

Comp3D

MMVII supports a subset of Comp3D *OBS* format (https://ignf.github.io/Comp3D/doc/obs.html).


Comp3D
Project Tools Comp3D

Final Points

	c	Name	Δx m	Δy m	ΔEh m
0	3xyz	1	0.0000	0.0000	-0.0000
1	3---	201	0.0000	0.0001	-0.0000
2	3---	202	0.0000	0.0000	0.0000
3	3---	204	-0.0001	0.0000	0.0000
4	3---	114	0.0000	0.0001	0.0000
5	3---	115	0.0000	0.0000	0.0000
6	3---	116	0.0000	0.0000	0.0000
7	3---	101	0.0000	0.0000	0.0000
8	3---	102	0.0000	0.0000	0.0000
9	3---	108	-0.0000	-0.0000	0.0000
10	3---	109	-0.0001	0.0000	0.0000
11	3---	110	-0.0001	0.0000	0.0000
12	3---	111	-0.0000	0.0000	0.0000
13	3---	112	-0.0000	0.0000	0.0000
14	3---	113	-0.0000	0.0001	0.0000

Final Observations

	Act	Origin	Target	Type	σ	Res σ
0	✓	1	1	crd_x	0.0000 m	
1	✓	1	1	crd_y	0.0000 m	
2	✓	1	1	crd_z	0.0000 m	
3	✓	1	201	tour	0.0008 g	0.9
4	✓	1	201	azim	0.0008 g	
5	✓	1	201	zen	0.0008 g	-0.4
6	✓	1	201	dist	0.0010 m	0.1
7	✓	1	202	hz	0.0008 g	-0.9
8	✓	1	202	zen	0.0008 g	0.3
9	✓	1	202	dist	0.0010 m	0.2
10	✓	1	204	hz	0.0008 g	-1.4
11	✓	1	204	zen	0.0008 g	-0.1
12	✓	1	204	dist	0.0010 m	0.7
13	✓	1	114	hz	0.0008 g	0.6
14	✓	1	114	zen	0.0008 g	-1.6

Configuration Reload Start Compensation Interrupt View Log

End of Computation after 2 iterations, final sigma0=1.16316. Duration: 00:00:00.034621

Comp3D is IGN's reference software for topometric computation.
Newly open source: https://ignf.github.io/Comp3D

MMVII

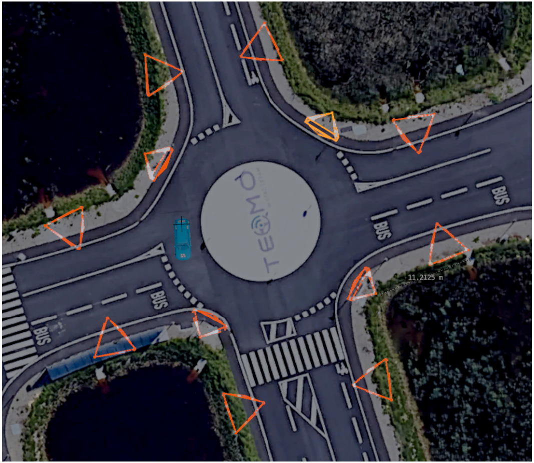
In its global compensation stage, **MMVII** can handle:

- Camera distortion models
- Tie points
- Ground control points (GCP)
- Rigid cameras blocks
- Clinometers (in progress)
- And topometric survey measurements!

Simulation

Project

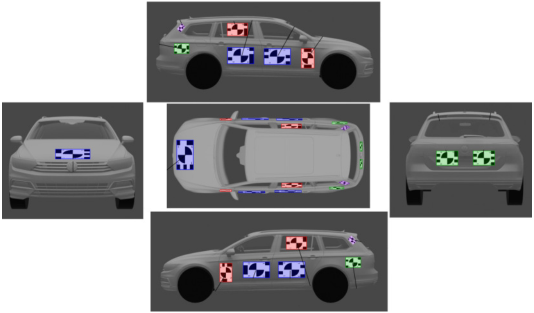
PRISSMA: follow a car around a round-about.



Project

Coded targets
Topometry
Simulation
▶ Project
Modelization
Computation
Conclusion

- 12 fixed known cameras, correct synchronisation
- Known ground points
- coded targets on car, known in car frame

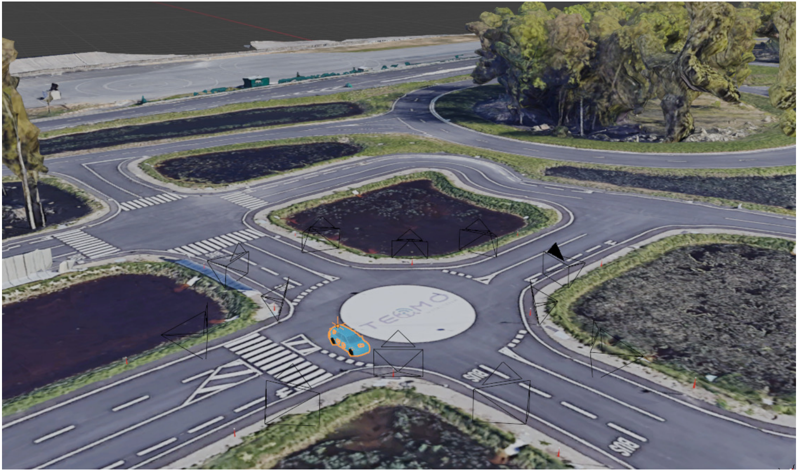


- Output trajectory: 5 Hz, 1 cm accuracy, $\approx$ 30 km/h

Simulation

- Optimize the cameras distribution
- Choose the focal lengths
- Distribute the ground reference points
- Optimize targets on the car (size / flatness / distribution)
- Simulate images with:
  - Camera resolution and noise
  - Depth of field
  - Motion blur

Blender Modeling

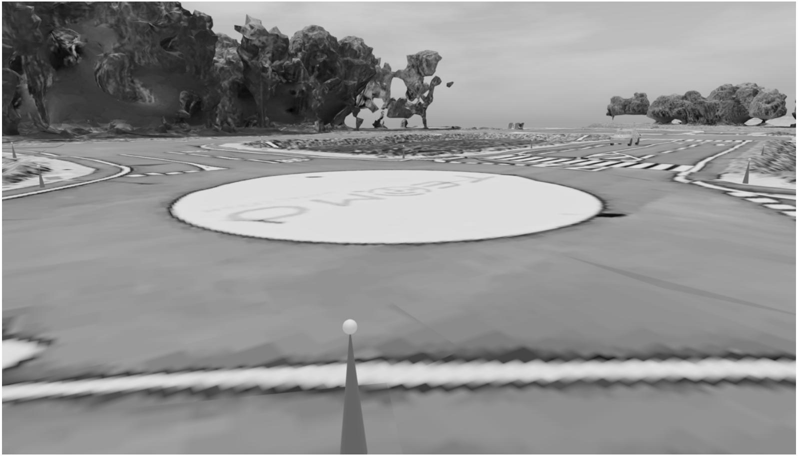


Export

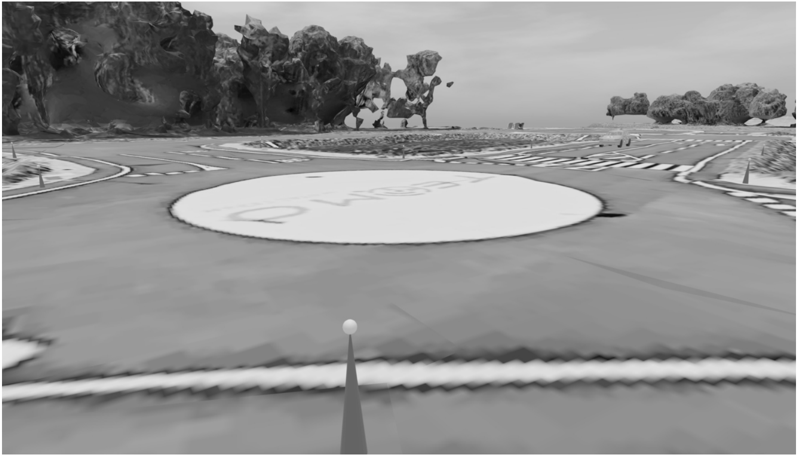
Blender Python API to export perfect parameters (+ noise):

- Camera calibration
- Image coordinates of the targets
- Ground coordinates of the targets
- Camera poses
- Car frame

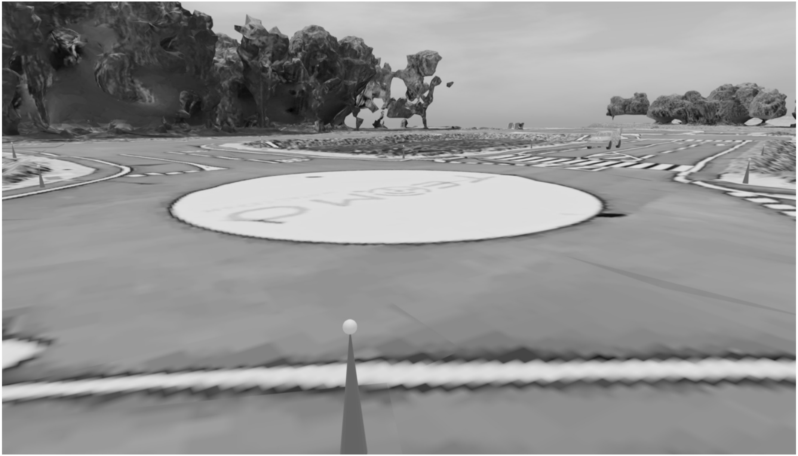
Rendering



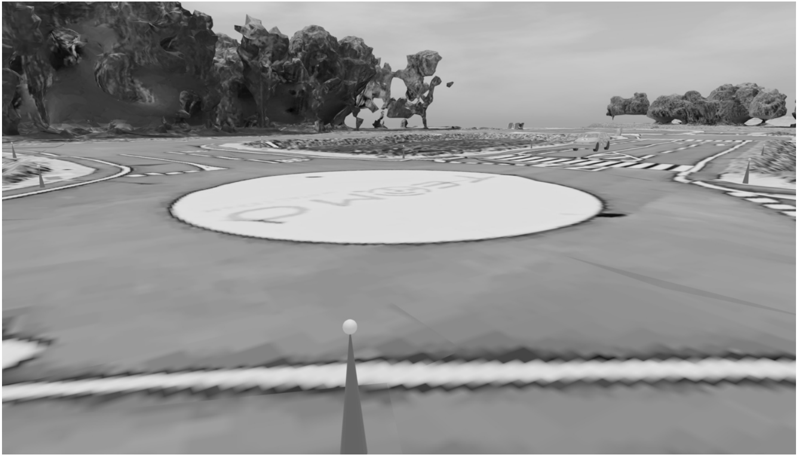
Rendering



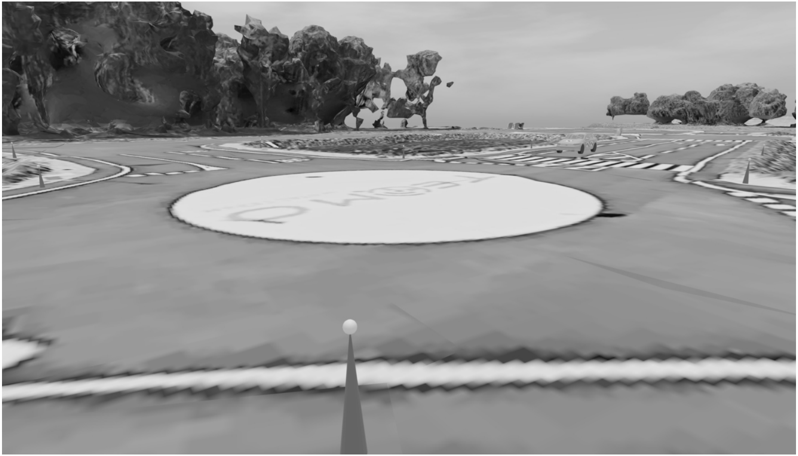
Rendering



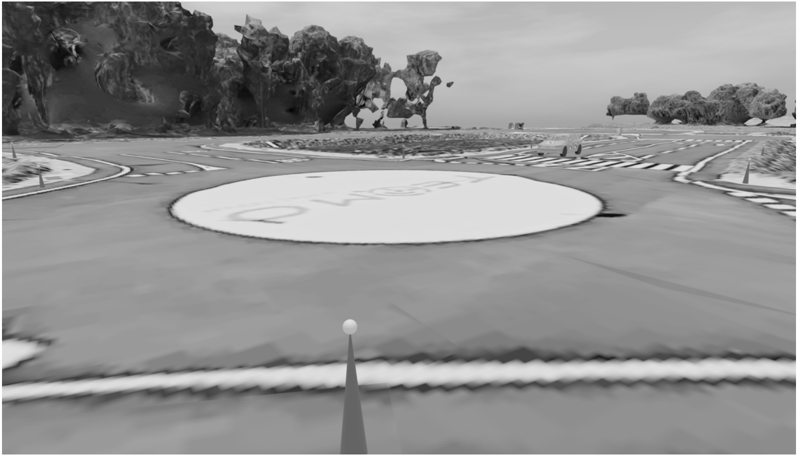
Rendering



Rendering

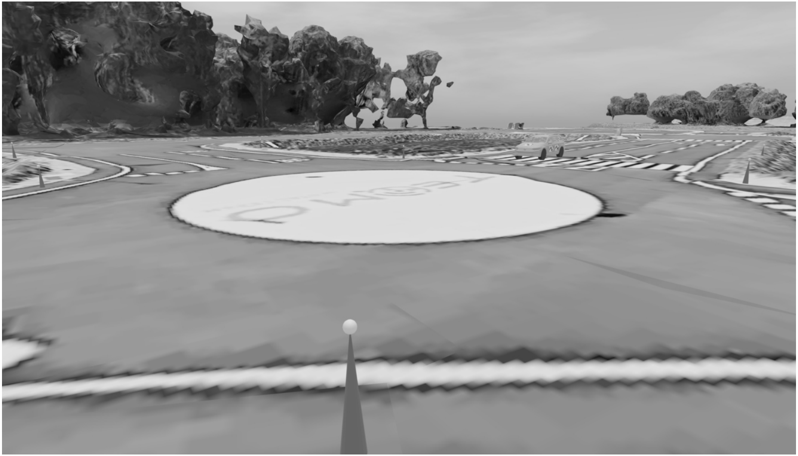


Rendering

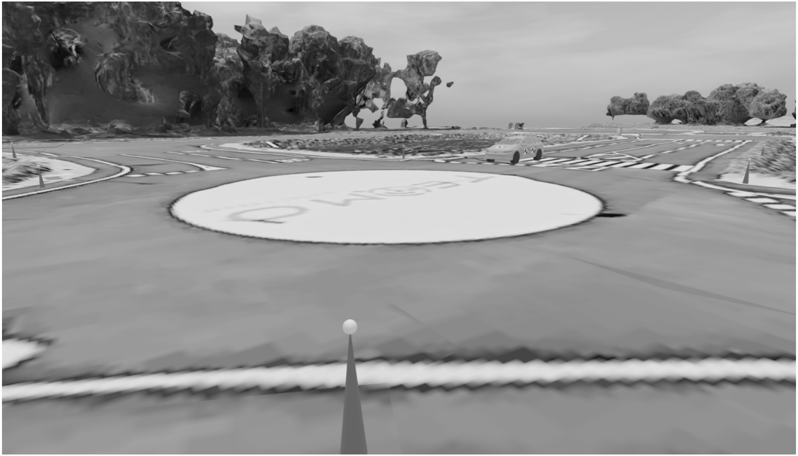


IGN

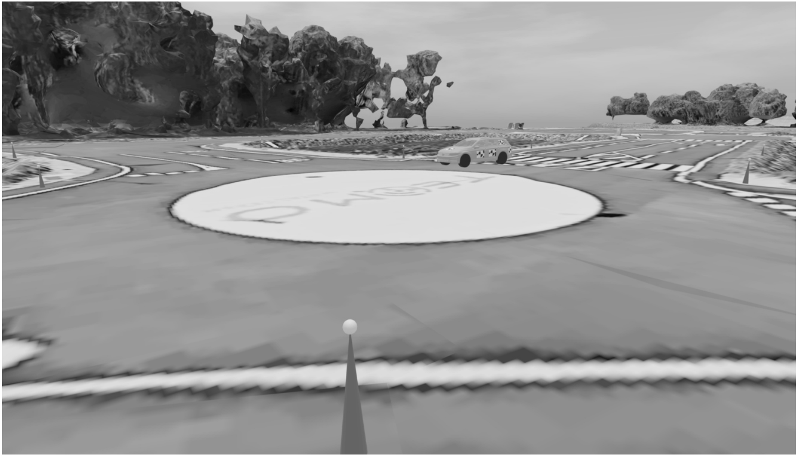
Rendering



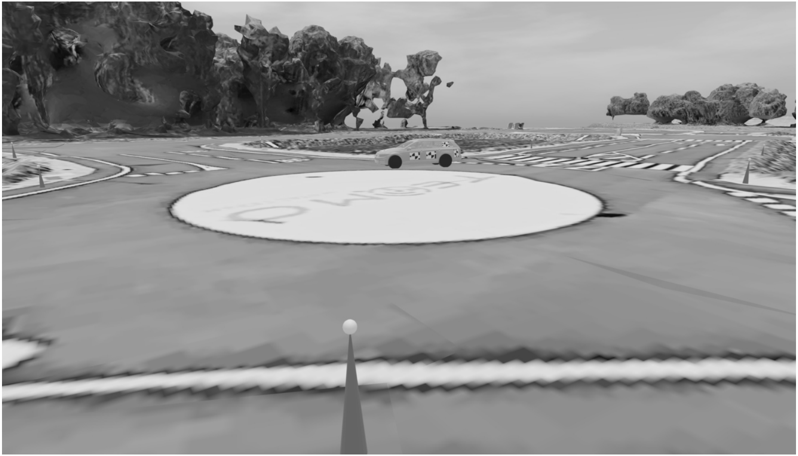
Rendering



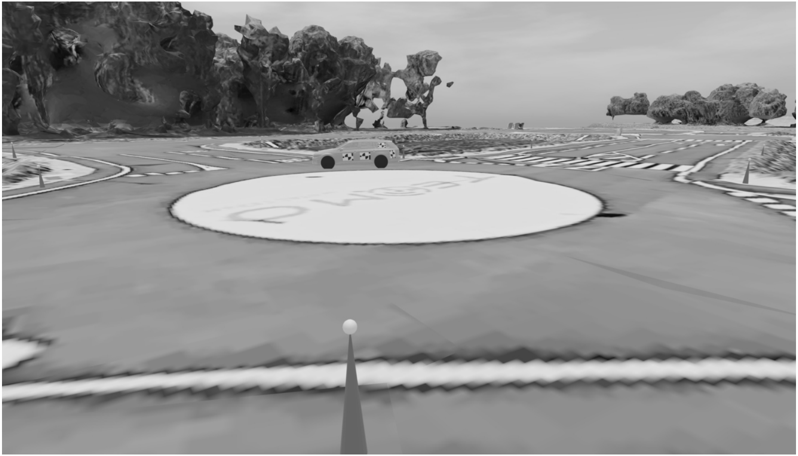
Rendering



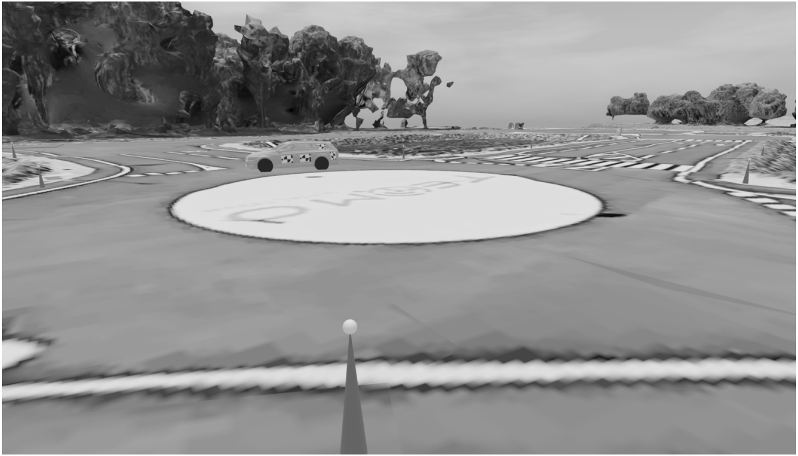
Rendering



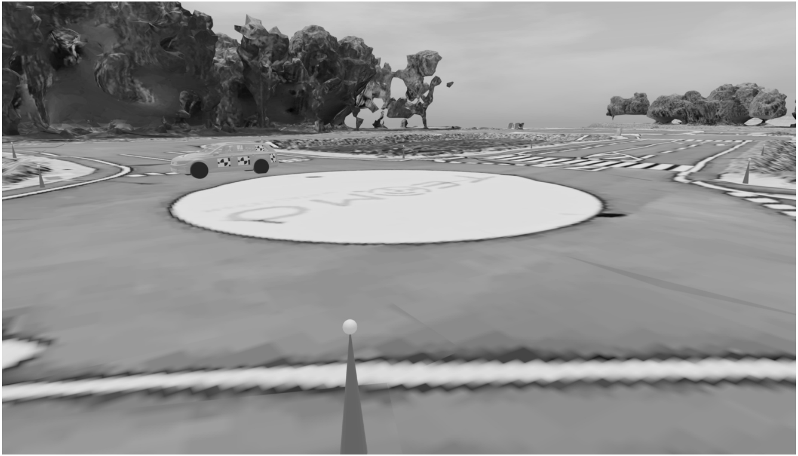
Rendering



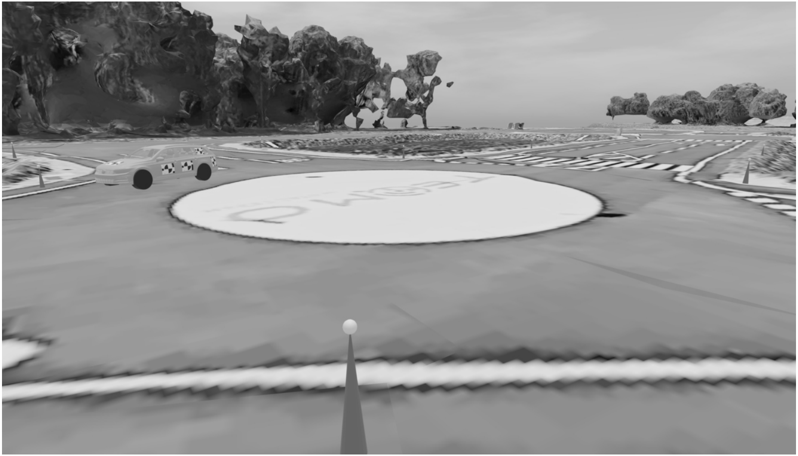
Rendering



Rendering



Rendering



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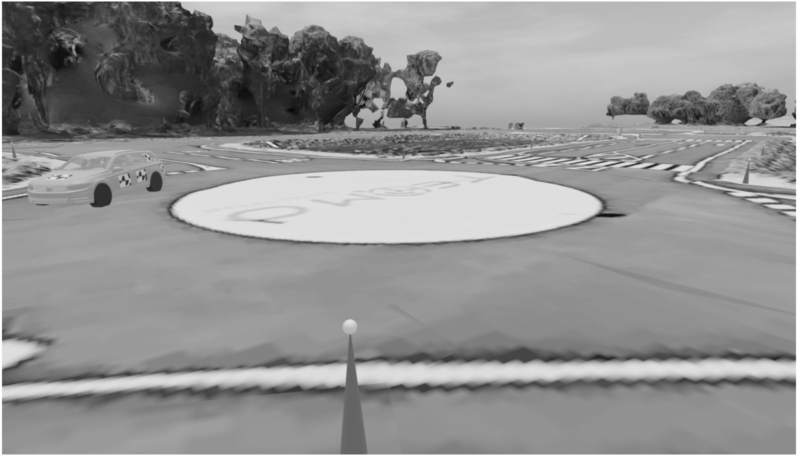
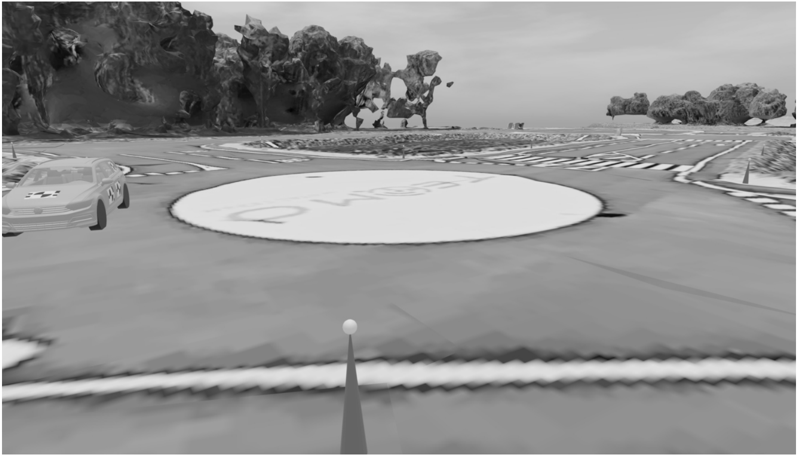
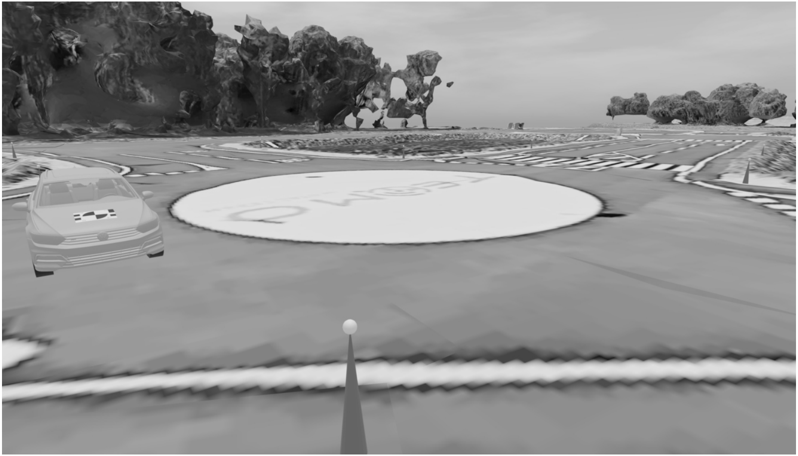


Rendering

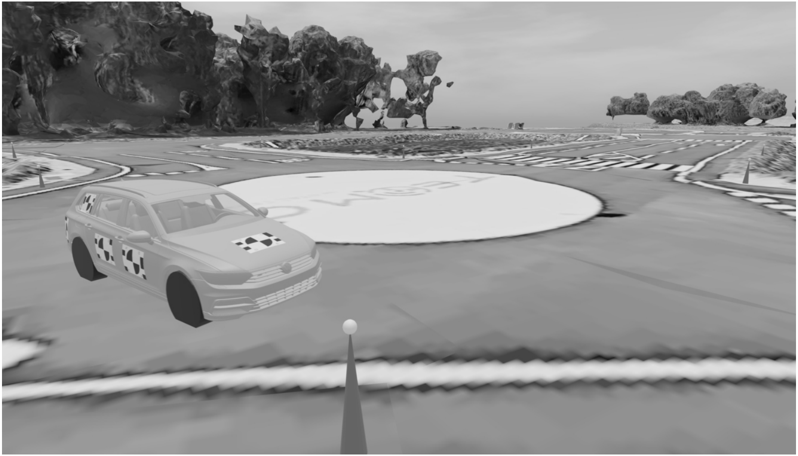


Rendering

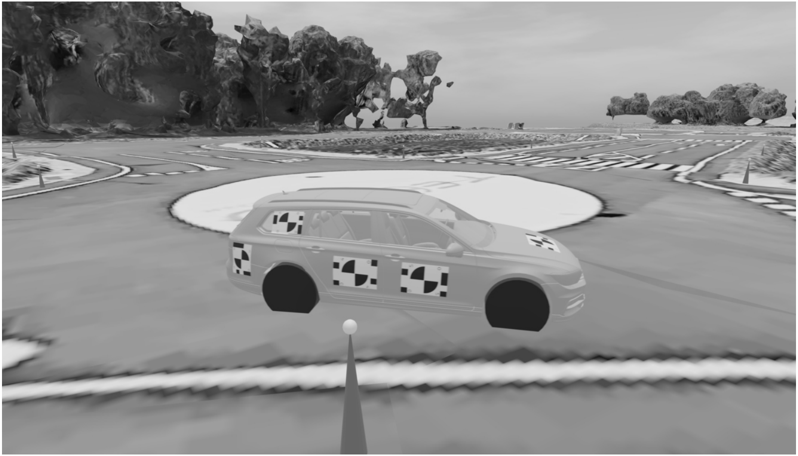


IGN

Rendering

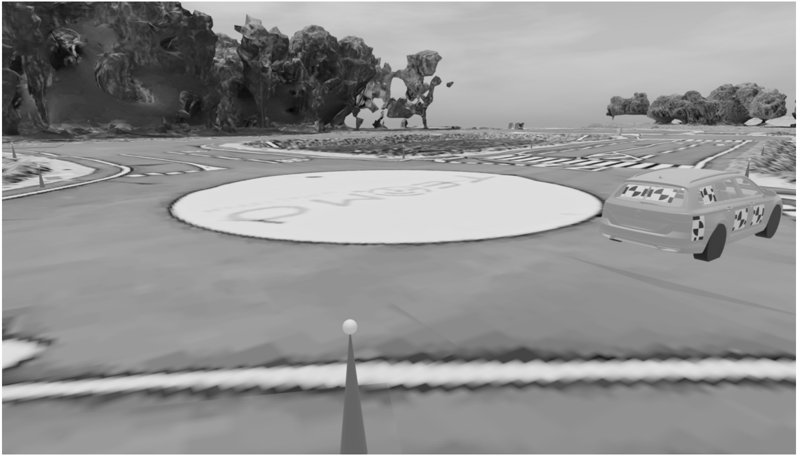


Rendering

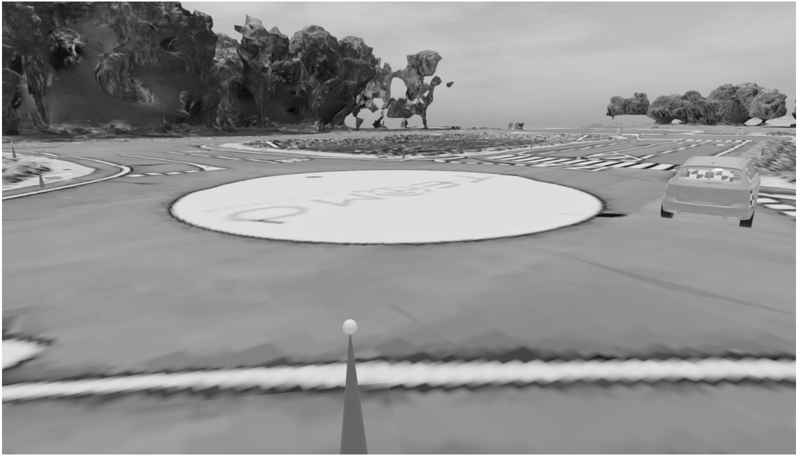


Rendering

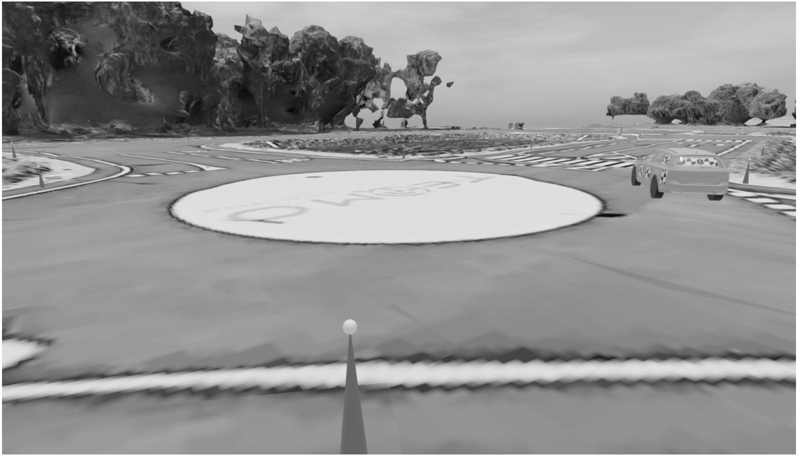
Rendering



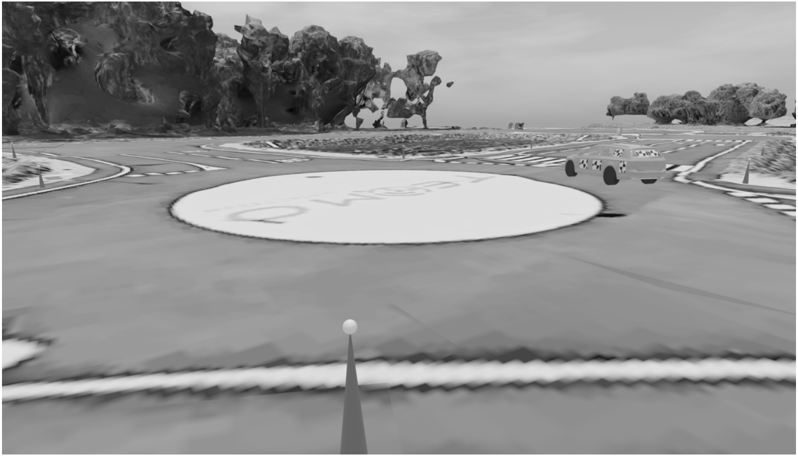
Rendering



Rendering

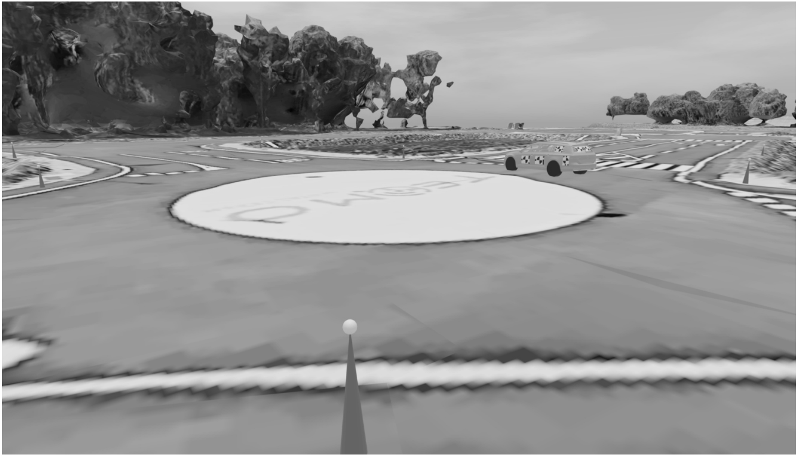


Rendering



IGN

Rendering



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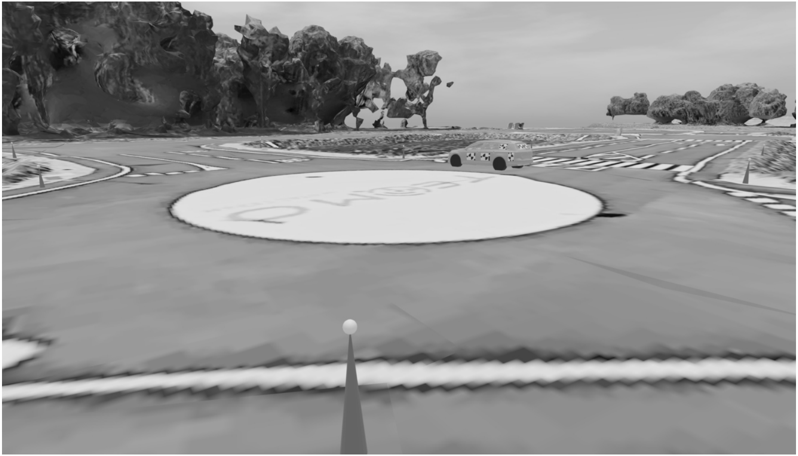

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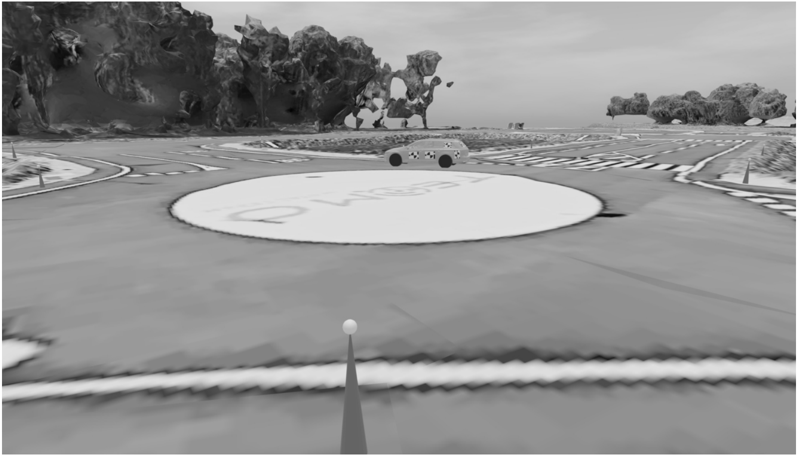

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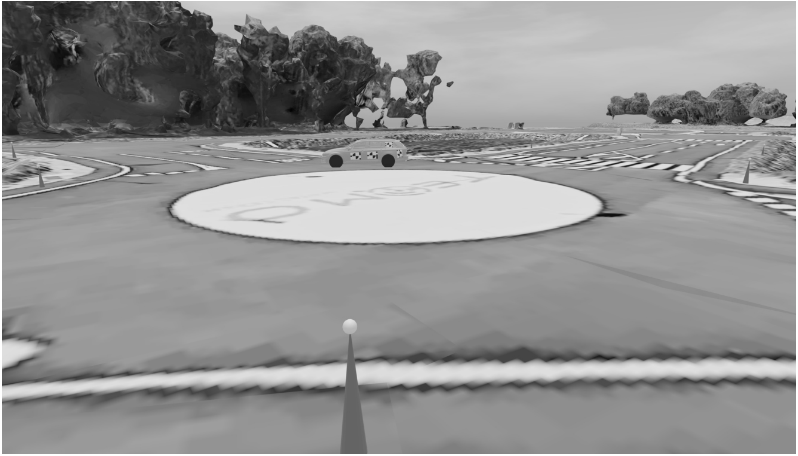

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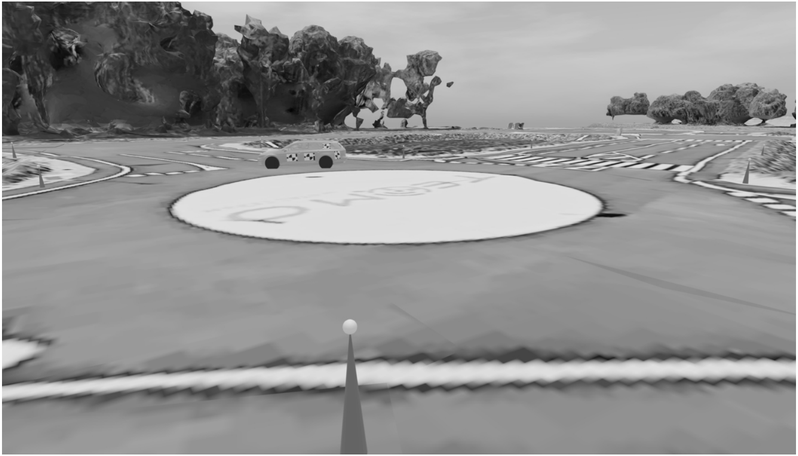

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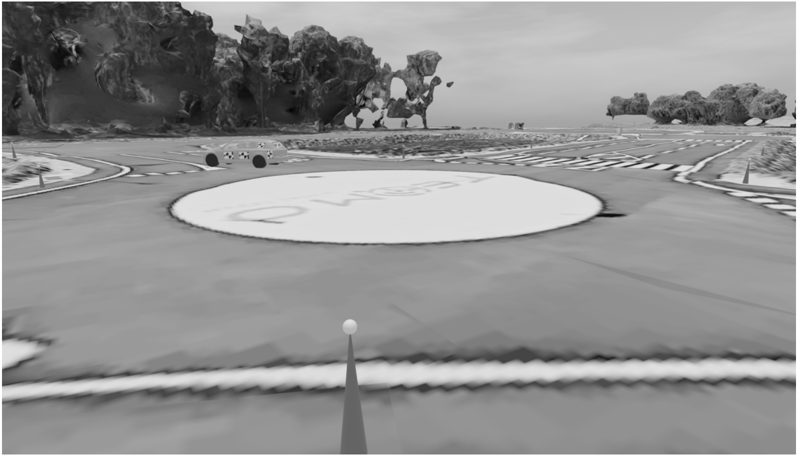

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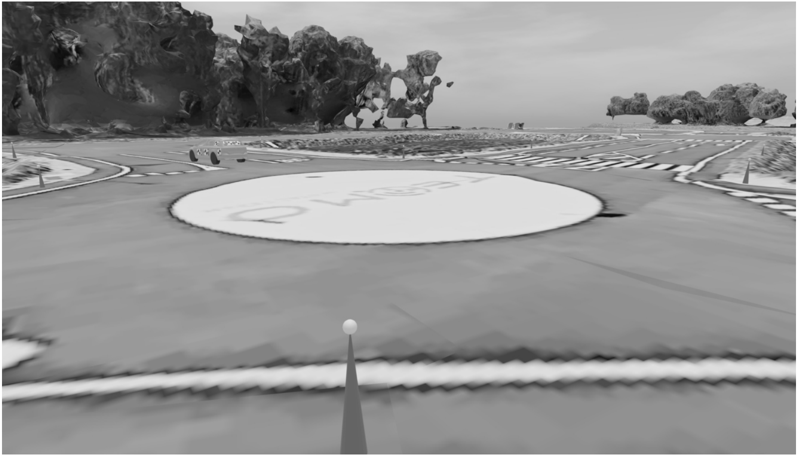

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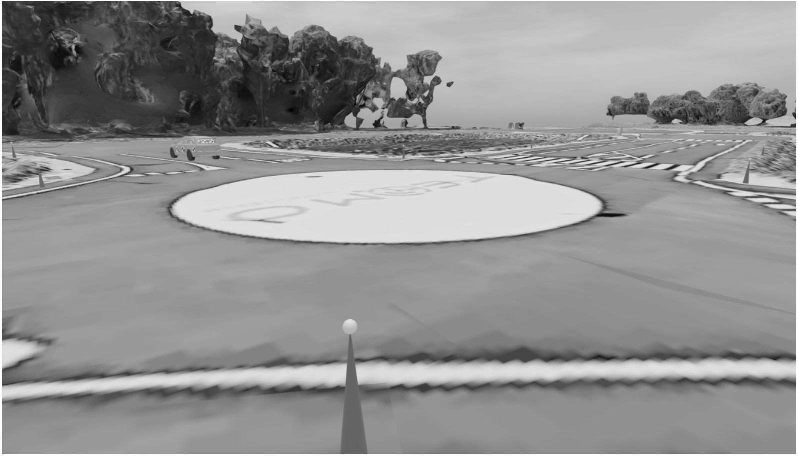
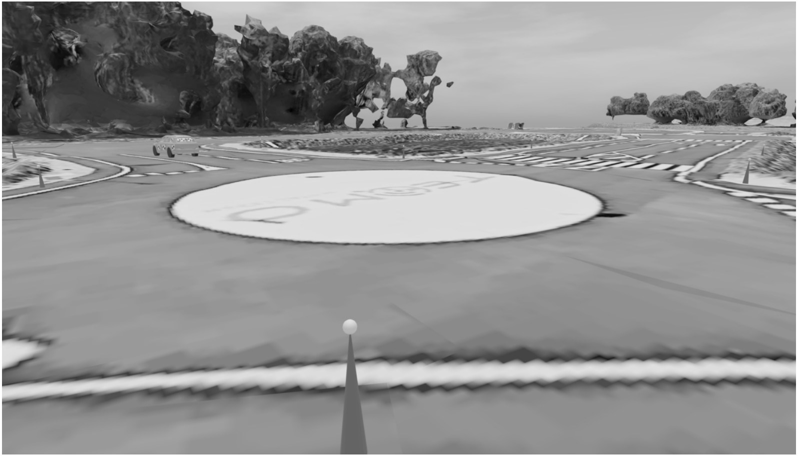


Rendering



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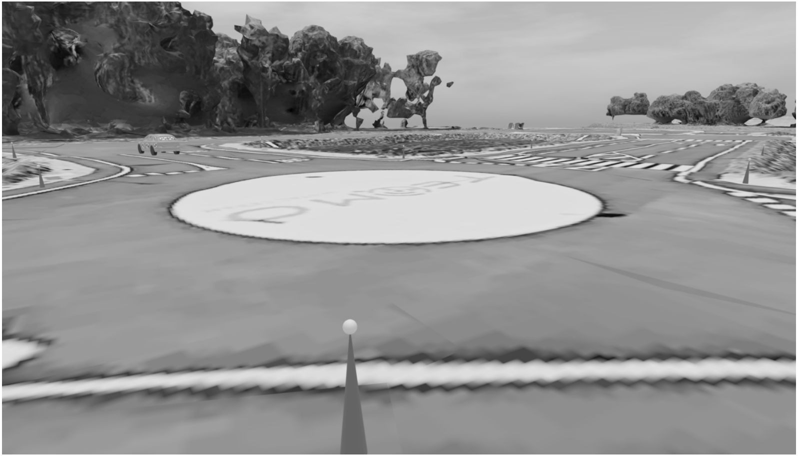

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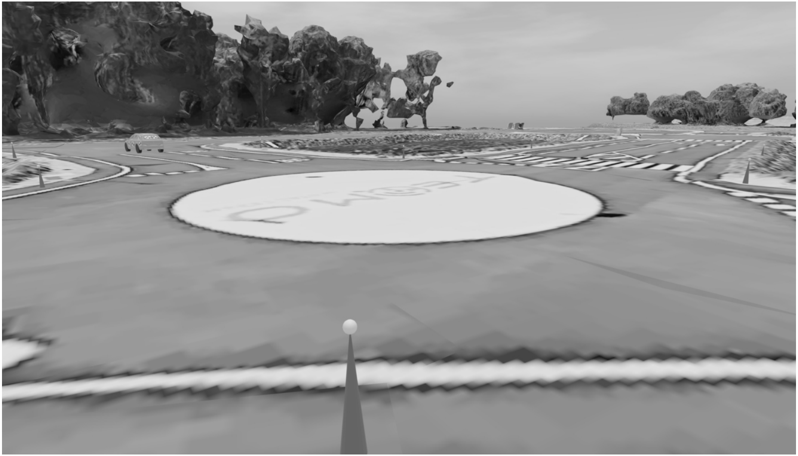

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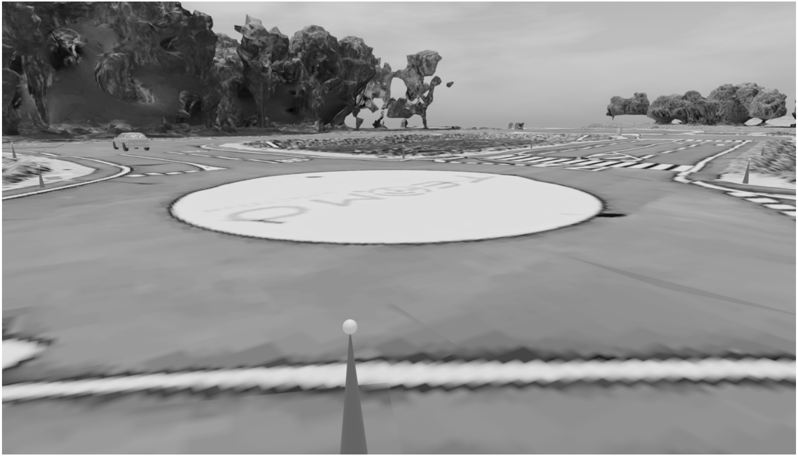
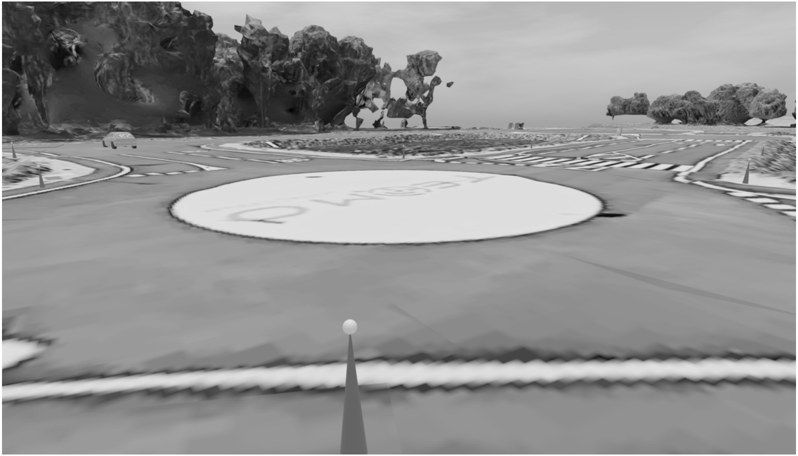
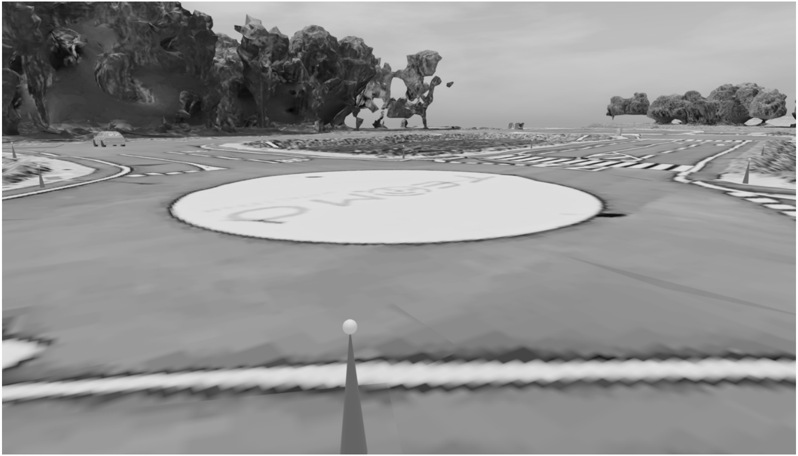


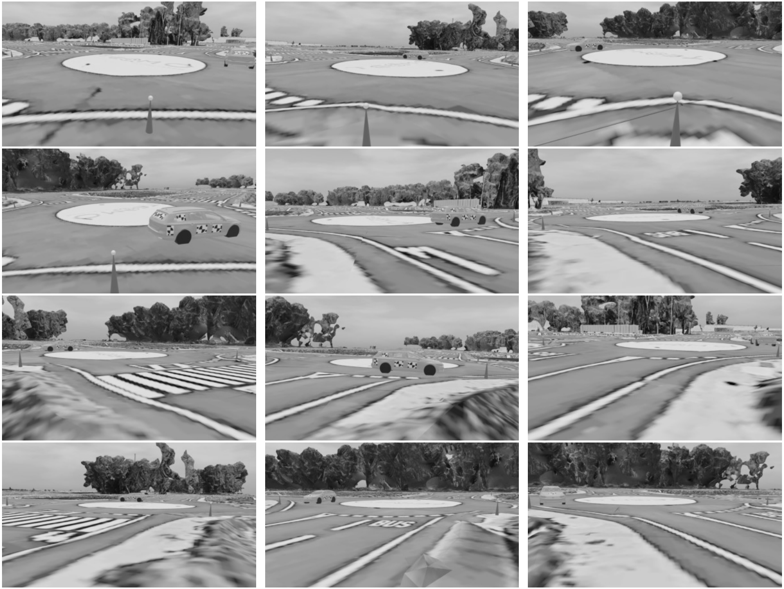
Rendering



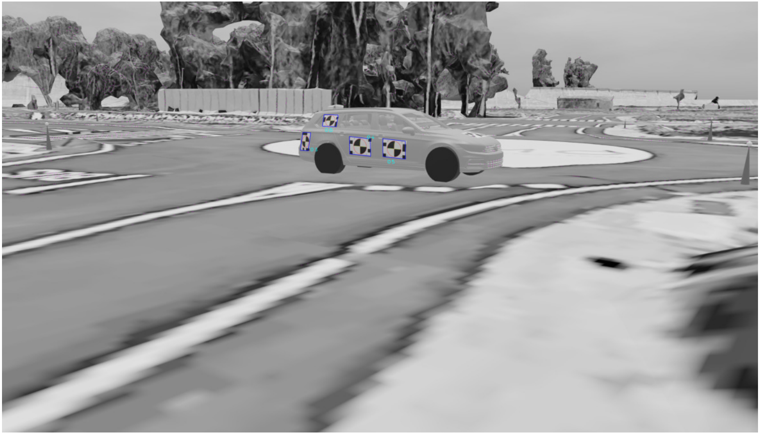
IGN

Rendering





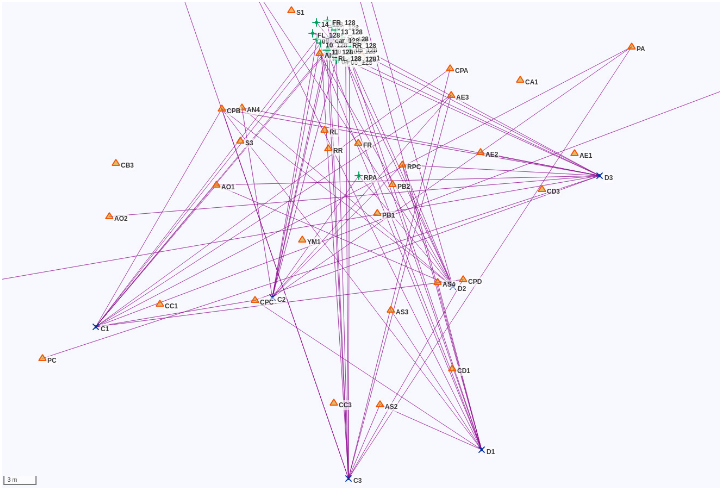
Target Detection



Comparable with the theoretical image coordinates exported by Blender.

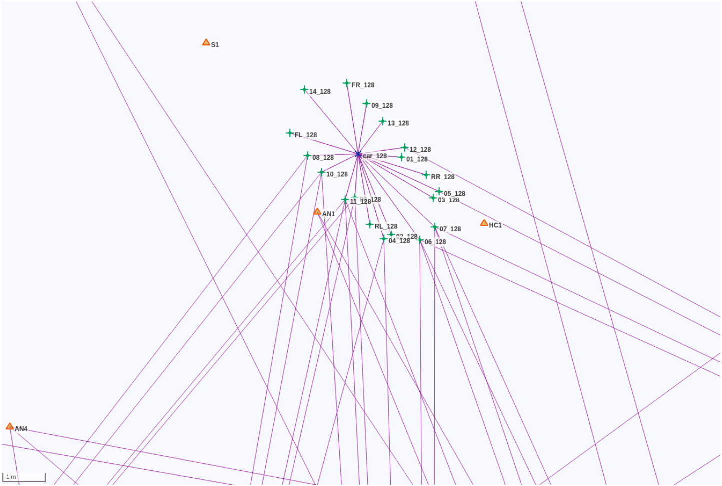
Adjustment

Build and run the entire computation chain (MMv2 + Comp3D):



IGN

Adjustment



S1
14_128
FR_128
09_128
13_128
FL_128
12_128
car_128
01_128
08_128
10_128
RR_128
05_128
03_128
11_128
AN1
HC1
RL_128
07_128
04_128
06_128
AN4
1 m

Adjustment

Output:

- Targets detection / decoding errors
- Trajectory coverage
- Compensated coordinates of the points and car frame
- Precision estimation
- Estimated vs theory

Adjustment

Different scenarios to test the system resilience:

- Noise on ground coordinates, car frame, targets or calibration
- Deactivation of some cameras
- Low light simulation (aperture, motion blur, noise)

Coded targets

Topometry

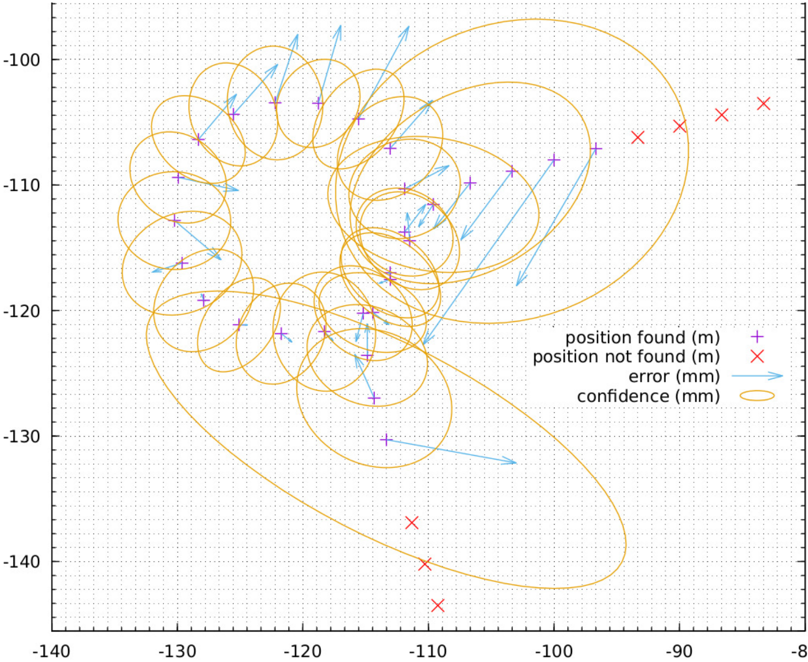
Simulation

Project

Modelization

▶ Computation

Conclusion



Acquisition

Acquisition

Follow-up

MicMac v2 evolutions:

- Topometry integration $\rightarrow$ the computation can now be entirely performed using MicMac v2
- Great improvement in target detection

Conclusion

Conclusion

MicMac v2 metrology computation improvements:

- Easy access to automatic measurements with subpixel precision
- Topometry integration
- Other types of constraints/measurements integration (rigid blocks, inclinometer)

Conclusion

https://github.com/micmacIGN/micmac/wiki/MMVII

https://github.com/micmacIGN/micmac/tree/master/MMVII

https://ignf.github.io/Comp3D